

100 Days of Azure DevOps

Day 44 Handout — Bicep is ARM with the JSON horror filed down — same types, same apiVersions, a compiler you install

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — a Bicep file is an ARM deployment waiting to compile

Bicep	Becomes	Notes
param location string = resourceGroup().location	ARM parameters	Default from the RG; still overridable
resource stg 'Microsoft.Storage/storageAccounts@2023-01-01'	resources[] with that apiVersion	Type string is the ARM type
sku: { name: 'Standard_LRS' }	properties/sku in JSON	Blocks instead of quote nests
name: 'st\${uniqueString(resourceGroup().id)}'	concat + uniqueString	Stops the landfill of colliding names
az bicep build -f main.bicep	main.json ARM	You can read the broccoli it emitted

Architecture B — if this is not your primary track

Track	Today's bar	Skip
Bicep primary	az bicep install; deploy storage; delete RG	Falling in love with JSON again
Terraform primary	30 min: install, type the resource block, build to JSON	Pretending Azure teams will never send .bicep
Both	Not a track	Pick Day 41's ADR

Architecture C — pitfalls

Pitfall	What you see	Fix
No compiler	az deploymentbicep fails oddly	az bicep install; az bicep version
Hardcoded stmyname123	AlreadyExists	uniqueString(resourceGroup().id)
location copied in six places	Second region = fork	param location
Deploy to a leftover RG	Complete-mode fear / name clashes	Fresh rg-day44-lab, Incremental

One-liner to remember

Bicep is ARM with the JSON horror filed down — same control plane, less eye strain. ARM did not go away.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. RG rg-day44-lab. Bicep track deploys; Terraform track still types the block once.

- 1 az bicep install && az bicep version. Confirm the CLI can compile.
- 2 Write infra/main.bicep from the starter. az bicep build -f infra/main.bicep and glance at the JSON once.
- 3 az group create -n rg-day44-lab -l centralindia. az deployment group create -g rg-day44-lab -f infra/main.bicep.
- 4 If Terraform is primary: stop after build + reading the resource symbol; optional deploy still recommended, then destroy.
- 5 Confirm the storage account exists (az resource list -g rg-day44-lab -o table).
- 6 az group delete -n rg-day44-lab --yes --no-wait.

main.bicep — storage with uniqueString

```
// infra/main.bicep
param location string = resourceGroup().location
param skuName string = 'Standard_LRS'

resource stg 'Microsoft.Storage/storageAccounts@2023-01-01' = {
  name: 'st${uniqueString(resourceGroup().id)}'
  location: location
  sku: { name: skuName }
  kind: 'StorageV2'
  properties: { allowBlobPublicAccess: false }
}

output storageName string = stg.name
output blobEndpoint string = stg.properties.primaryEndpoints.blob

# az bicep install
# az deployment group create -g rg-day44-lab -f infra/main.bicep
```

Done checklist

- az bicep version works
- Resource type string includes apiVersion
- Name uses uniqueString, not pride
- RG deleted after proof

Tomorrow — Day 45

Bicep modules and what-if — split storage out, read Create/Modify/Delete before the audience arrives.